



Assignment 02

Natural Language Processing

Submitted by: Aleena Zahra

Roll number 23i-2514

Date: 19 April, 2026



National University of Computer and Emerging Sciences Islamabad Campus

Overview

This project is about using computers to understand and work with the Urdu language. I focused on three areas: word embeddings, sequence tagging and text classification. the goal was to see how well different models and techniques work for Urdu especially when there is not data available. I used a dataset of articles to test the models.

Part 1 Word Embeddings

I created representations of words using Term-Frequency Inverse Document Frequency (TF-IDF) and Positive Pointwise Mutual Information (PPMI). I also trained a Skip-gram Word2Vec model. I checked how good these embeddings were by looking at neighbors and doing analogy tests.

- * MRR Comparison: The Skip-gram Word2Vec embeddings worked better than the PPMI baseline. Here are the scores:
- * C1. PPMI baseline: 0.0423
- * C2. Skip-gram on raw.txt: 0.0610
- * C3. Skip-gram on cleaned.txt (d=100): 0.0592
- * C4. Skip-gram on cleaned.txt (d=200): 0.0560
- * Qualitative Assessment: I looked at the neighbors of words. Did analogy tests. For example Pakistan was close to Afghanistan and India. This shows that the model understands geography and politics.

Part 2 Sequence Tagging

I built a model to tag parts of speech (POS). Named entities (NER). I used a 2-layer Bidirectional Long Short-Term Memory (BiLSTM) network. I also used rules and dictionaries to help the model.



National University of Computer and Emerging Sciences Islamabad Campus

- * POS Tagging: the BiLSTM model for POS tagging was not very accurate. It got an accuracy of 0.6272 and a Macro-F1 score of 0.1101. I looked at the mistakes. Saw that 'POST' 'VERB' and 'PRON' were often mislabeled as 'NOUN'.
- * NER Tagging: the NER model did not work all. It got a Macro-F1 score of 0.0000. This means I need to work on the dataset or model.
- * Ablation Study: I tested versions of the POS tagging model.
- * A1 (Unidirectional LSTM): This version did worse showing that looking at context in both directions is important.
- * A2 (No Dropout): This version did suggest that the model was overfitting.
- * A3 (Random Embeddings): This version did poorly showing that pre-trained Word2Vec embeddings are helpful.
- * A4 (Softmax NER): This version did better than the NER model but still not great.

Part 3 Transformers

I built a Transformer-based classifier to classify articles into topics.

The model learned quickly because the classification problem was too easy.

Attention Heatmaps: I looked at which tokens the model focused on when making decisions. This helps us understand how the model works.

Conclusion

This project shows that working with Urdu language data can be challenging. The word embeddings worked well. The sequence tagging models, NER did not. The Transformer classifier showed that I need a more diverse dataset.